

SEDS519-HW2/Interactive Mailing System

Definition:

For your second homework, you will be writing a program for an interactive mailing service. It will contain the basic principalities of mailing service with the addition of interactive file handling. This interactiveness comes from the ability of the user to convert a given type of a file into different types of different file extensions, while also preserving the structure. The users of the system will also have the abilities of displaying, updating, converting, and etc. of attachments.

Objectives:

1. **(30 pts) Model-View-Controller (MVC):** The entire application must be implemented with compliance to the MVC pattern. The model part is realised via mail and task data structures. The view is represented via GUI for displaying the inbox and a given mails' content and attachment visualisation. Lastly, the Controller will be responsible for the organisation of user actions. The actions such as 'Reply', 'Update', and 'Convert'. These three layers must remain loosely coupled.
2. **(30 pts) Adapter Pattern (File Conversion):** The adapter pattern will be applied in the implementation of conversion of different file formats (PDF, CSV, Excel) into standard HTML table format that enables display and in-place editing. Each file type should have its own Adapter class implementing a common FileAdapter interface.
3. **(20 pts) Composite Pattern (Mail Structure):** Since each mail may have differing format of contents, such as one mail having only text and prompts the receiver to send a file whilst the other may have a semi filled attachment that alongside the informative texts, and prompts the receiver to fill the required information on the attachment etc. The composite pattern will be applied for treating individual attachments and groups of attachments uniformly within a single mail object. Define a MailComponent interface with Leaf (single file) and Composite (mail with multiple parts) implementations.

Notes:

- **(10 pts)** You should check the validity of the mail (whether it has an empty body or no attachments).
- **(5 pts)** You should create a GUI that lets the user display both the mail and attachments.
- **(5 pts)** You should draw Unified Modeling Language (UML) class diagrams for each class, taking into account class interactions.